EA-MT

Entity-Aware Machine Translation in English to Chinese

Jou-Yi Lee, Camellia Bazargan, Yifei Gan

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Contents

1 – Introduction

2 – IWSLT En-Zh Dataset

3 – EA-MT Flowchart

4 – NER - SpaCy

5 – mBART

6 – EA-MT

7 – Evaluation Flowchart

8 – METEOR & JieBa

9 – HIT-CIR (Tongyici Cilin 同義詞詞林) WordSimilarity Dataset & OpenHowNet

10 – Evaluation Results

PART.01

Introduction

EA-MT : Entity-Aware Machine Translation

English to Chinese

Introduction Section

Challenges in NER

- **Contextual Ambiguity :**
  Apple (Fruit? Or Company?)
- **Multi-word Entities:**
  literal translations often produce awkward outputs. Department of Justice 司法部
- **Cross-lingual Consistency :**
  Entities can have multiple valid translations

Study Focus

- **IWSLT 2017 NER Translation Task**
- **SpaCy for NER**
- **mBART**
  Integrates these entities into a machine translation model
- **Enhance Translation Accuracy**

Objective

- Named Entity Translation Model
- Maintain context and semantic accuracy when doing cross-lingual machine translation jobs.

Motivation

The inherent difficulty of this task was interesting to us, as translating sentences containing named entities are notoriously difficult due to linguistic and cultural differences. They often need specialized attention, meaning that traditional MT models are usually too general to translate these entities correctly.

We've chosen Chinese since not only does two of our three team members speak it and could vouch for the accuracy of the results, but also because English and Chinese are as different as two languages could be, so it would be interesting to observe what sorts of translations could be the result.

Related works

Tang, Y., Tran, C., Li, X., Chen, P., Goyal, N., Chaudhary, V., Gu, J., & Fan, A. (2020).
Multilingual Translation with Extensible Multilingual Pretraining and Finetuning. ***ArXiv, abs/2008.00401*****.**

Yinhan Liu, Jiatao Gu, Naman Goyal, Xian Li, Sergey Edunov, Marjan Ghazvininejad, Mike Lewis, Luke Zettlemoyer;
Multilingual Denoising Pre-training for Neural Machine Translation. Transactions of the Association for Computational Linguistics 2020; 8 726–742. doi: https://doi.org/10.1162/tacl_a_00343

Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2019).
BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. North American Chapter of the Association for Computational Linguistics.

Xie, S., Xia, Y., Wu, L., Huang, Y., Fan, Y., & Qin, T. (2022).
End-to-end entity-aware neural machine translation. Machine Learning, 111, 1181 - 1203.

Pelofske, E., Urias, V., & Liebrock, L. M. (2024, April 23).
Automated Multi-Language to English machine Translation using Generative Pre-Trained Transformers. arXiv.org. https://arxiv.org/abs/2404.14680

PART.02

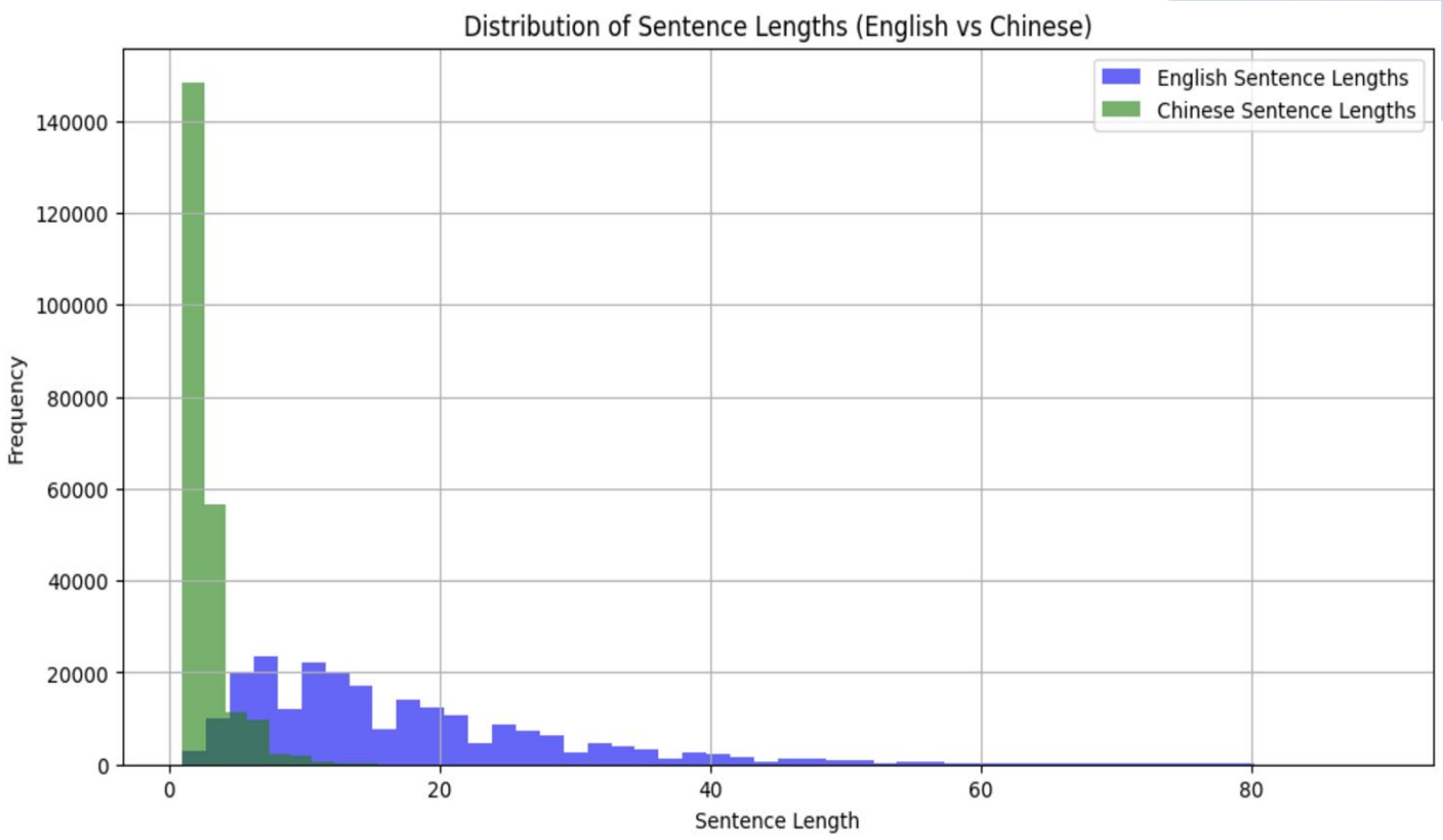
IWSLT En-Zh Dataset

IWSLT

IWSLT 2017 En-Zh

- IWSLT is Real-World conversation dataset.
- English sentences are longer than Chinese translations.
- total count of 240694 samples.

The Chinese language is character-based.



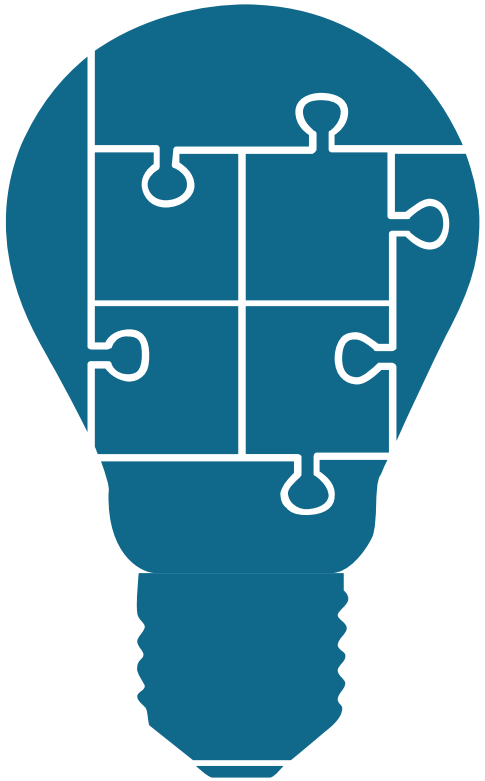
Linguistic Structural Differences

Character-Based vs. Word-Based

Chinese uses characters as the smallest semantic units.

Example:

[The dog is on the mat.]
(6 English words)
[“狗” ,”在”, ”墊子”, ” 上”]
(5 Chinese characters)



Word Order and Syntax

- English follows a strict Subject-Verb-Object (SVO)
- Chinese allows flexible structures.

Example:

[I like this book]
Standard Chinese
[我喜歡這本書]
Emphasis:
[這本書，我喜歡]

These will cause inaccuracies in translations.

Word Cloud Analysis

Common function words
(e.g., "the," "and," "to ,"of")

High-frequency tokens
(“的,”“是,”“我,“)

Conform to Zipf's Law.

Sentence Type Analysis

Complex Sentences:

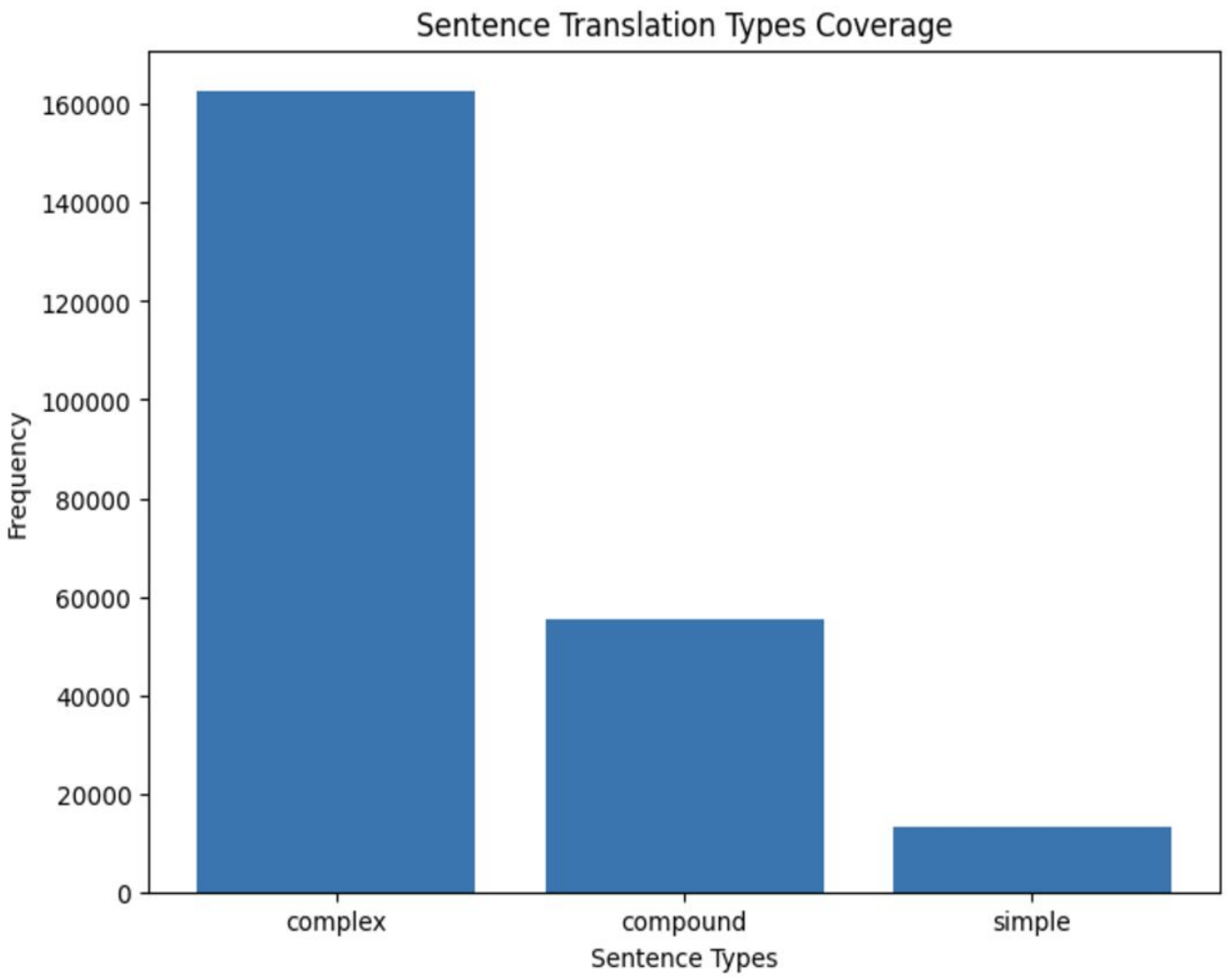
Have “because,” “although”

Compound Sentences:

Have "and" or "but”

Implications for Machine Translation:

1. Character-Based vs. Word-Based
2. Word Order and Syntax
3. Zipf's Law
4. Different Sentences Type

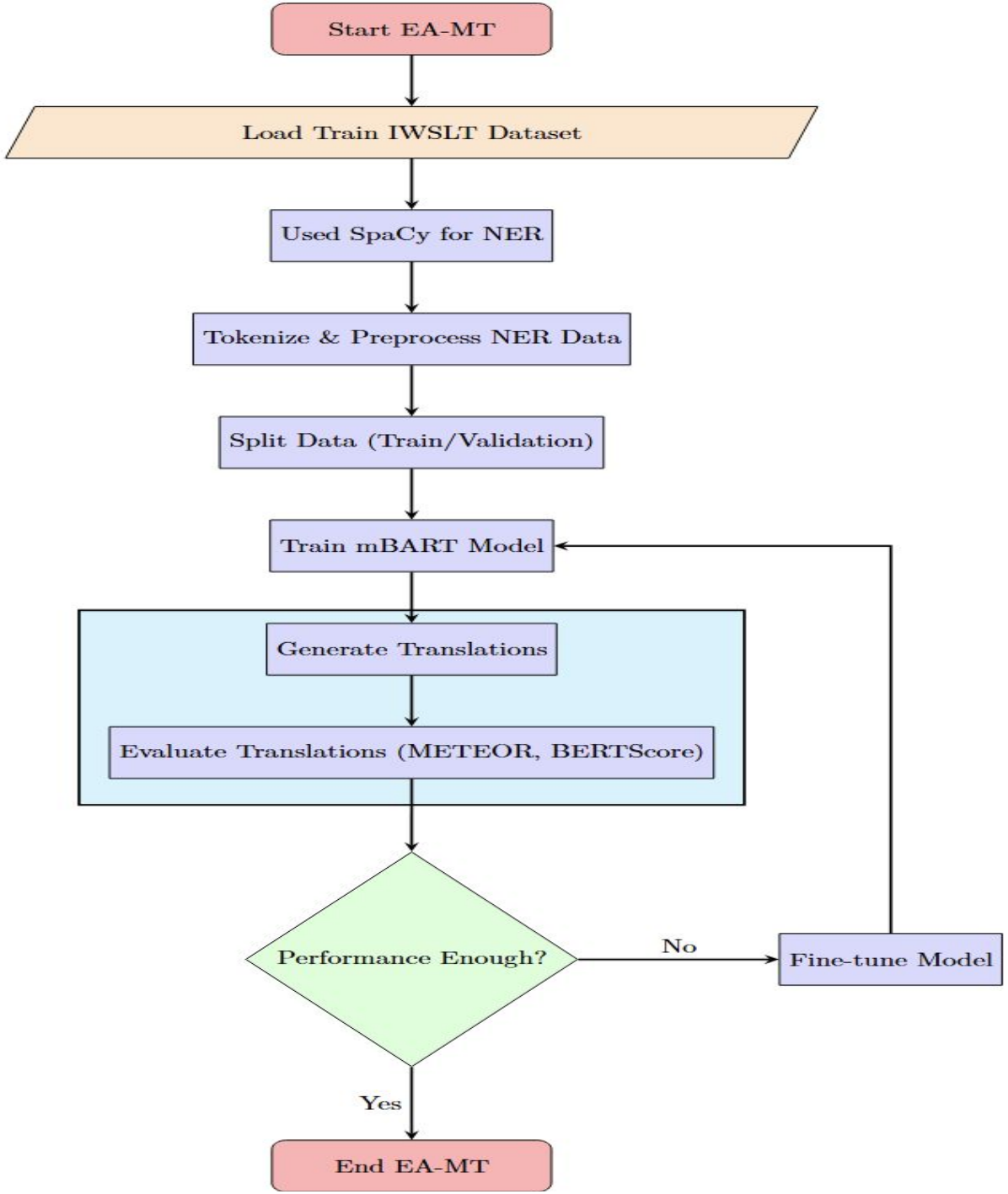


PART.03

EA-MT Flowchart

Entity-Aware Machine Translation Flowchart

EA-MT Flowchart



PART.04

NER - SpaCy

Named Entity Recognition using

Named Entity Translation Process

SpaCy:

1. Transformer-based NER model
2. Used (en_core_web_trf) model
3. **<<Entity Label : Entity Text>>**
   Improved the model's awareness of named entities
   <<ORG : Apple>>

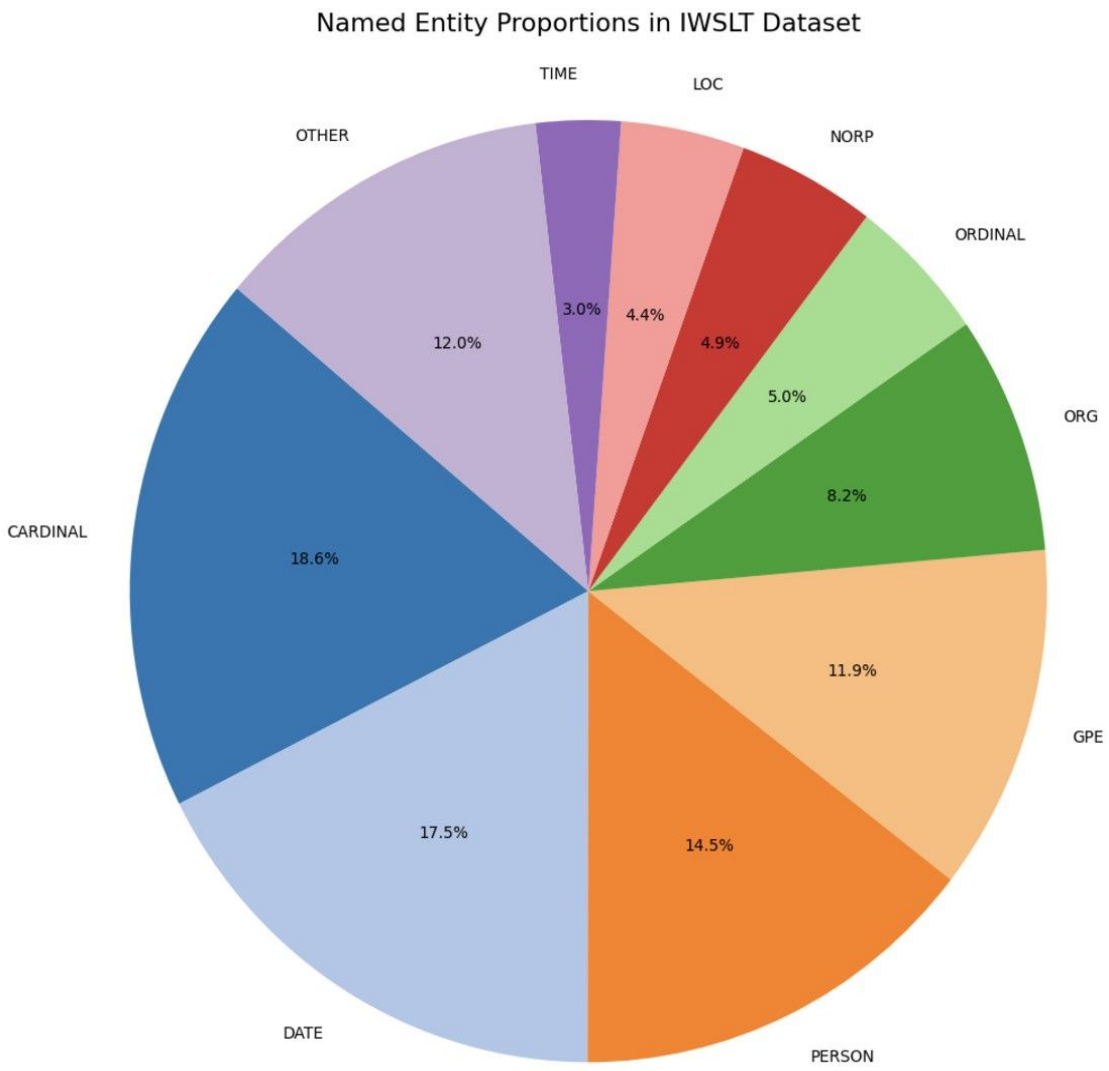
highlight the entities by a unique form in each sentence, let the model learn them itself during process.

compare to standfordNER/hanlp, more tags.

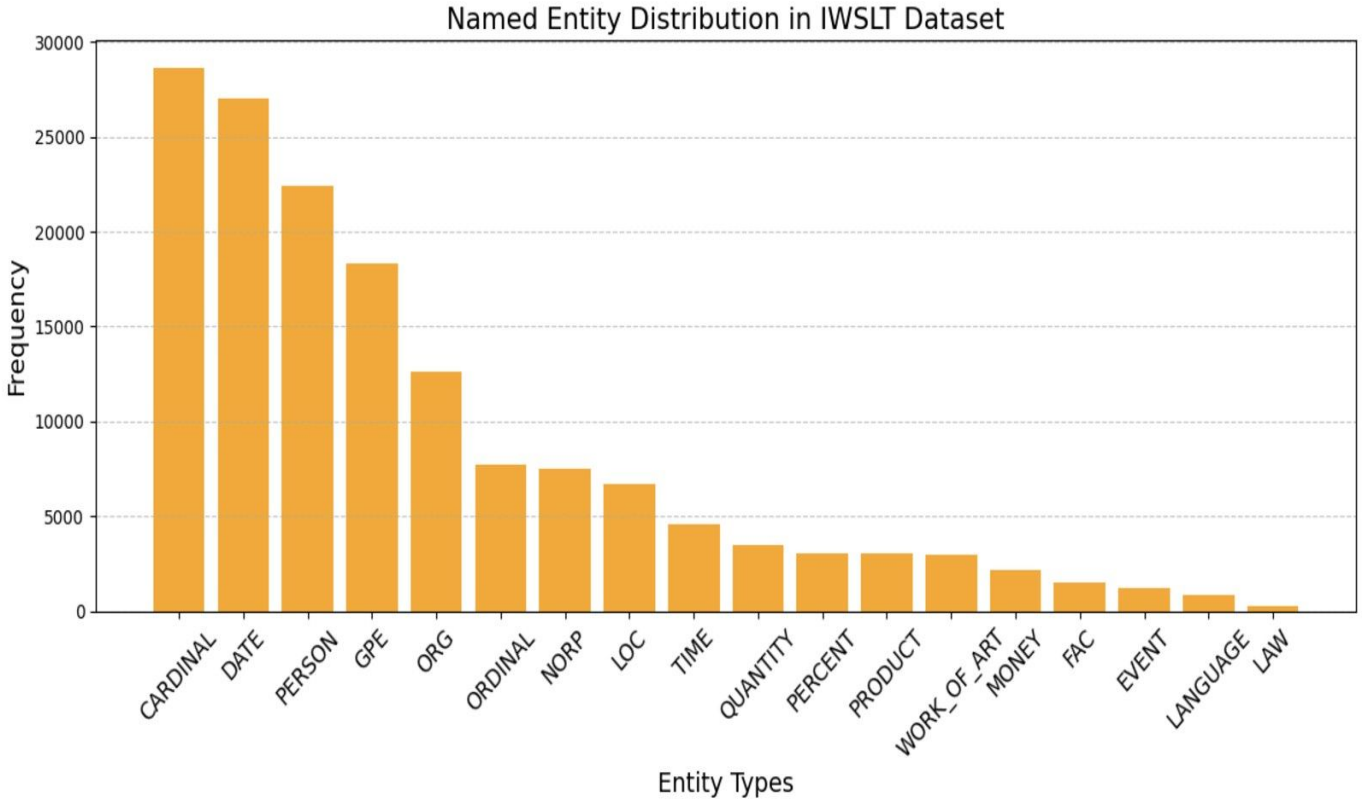
SpaCy: Have 18 different Tags

Tag	Description	Example
PERSON	A person's name	Barack Obama
NORP	Nationalities, religious groups, or political groups	American
FAC	Buildings, landmarks, or facilities	Eiffel Tower
ORG	Organizations, companies, or institutions	Google
GPE	Countries, cities, or geopolitical regions	France
LOC	Non-GPE locations (e.g., mountains, bodies of water)	Mount Everest
PRODUCT	Objects, vehicles, or products	iPhone
EVENT	Named events	Olympics
WORK_OF_ART	Titles of books, songs, or artworks	Mona Lisa
LAW	Legal documents or clauses	First Amendment
LANGUAGE	Languages	English
DATE	Dates or time periods	2024-01-01, Wednesday
TIME	Specific times	2 PM, noon
PERCENT	Percentage expressions	50%
MONEY	Monetary amounts	$20
QUANTITY	Measurements	10 kg
ORDINAL	Ordinal numbers	first, 2nd
CARDINAL	Cardinal numbers	one, 100

Visualization of NER Tag Distributions



- CARDINAL and DATE tags account for 36.1% of all named entities.
- Entities are mainly distributed in 10 categories.
- Base on this we can adjust the weight of the model.



PART.05

mBART

mbart-large-50-many-to-many-mmt from Meta AI

mBART

Architecture

- Encoder-Decoder Architecture
- Developed by Facebook AI

Model

- mbart-large-50-many-to-many-mmt
- 12 layers for both encoder and decoder
- Hidden dimension: 1024

Model Comparing

- mBERT:
  - bidirectional transformer model
  - Understanding tasks & question answering
- mBART:
  - Good at multilingual sequence-to-sequence tasks
  - mBART25 => 25 Languages
  - **BLEU** improvement 12 points

Configurations of mBART50

- Many-to-One (N→1)
- One-to-Many (1→N)
- **Many-to-Many (N↔N)**

PART.06

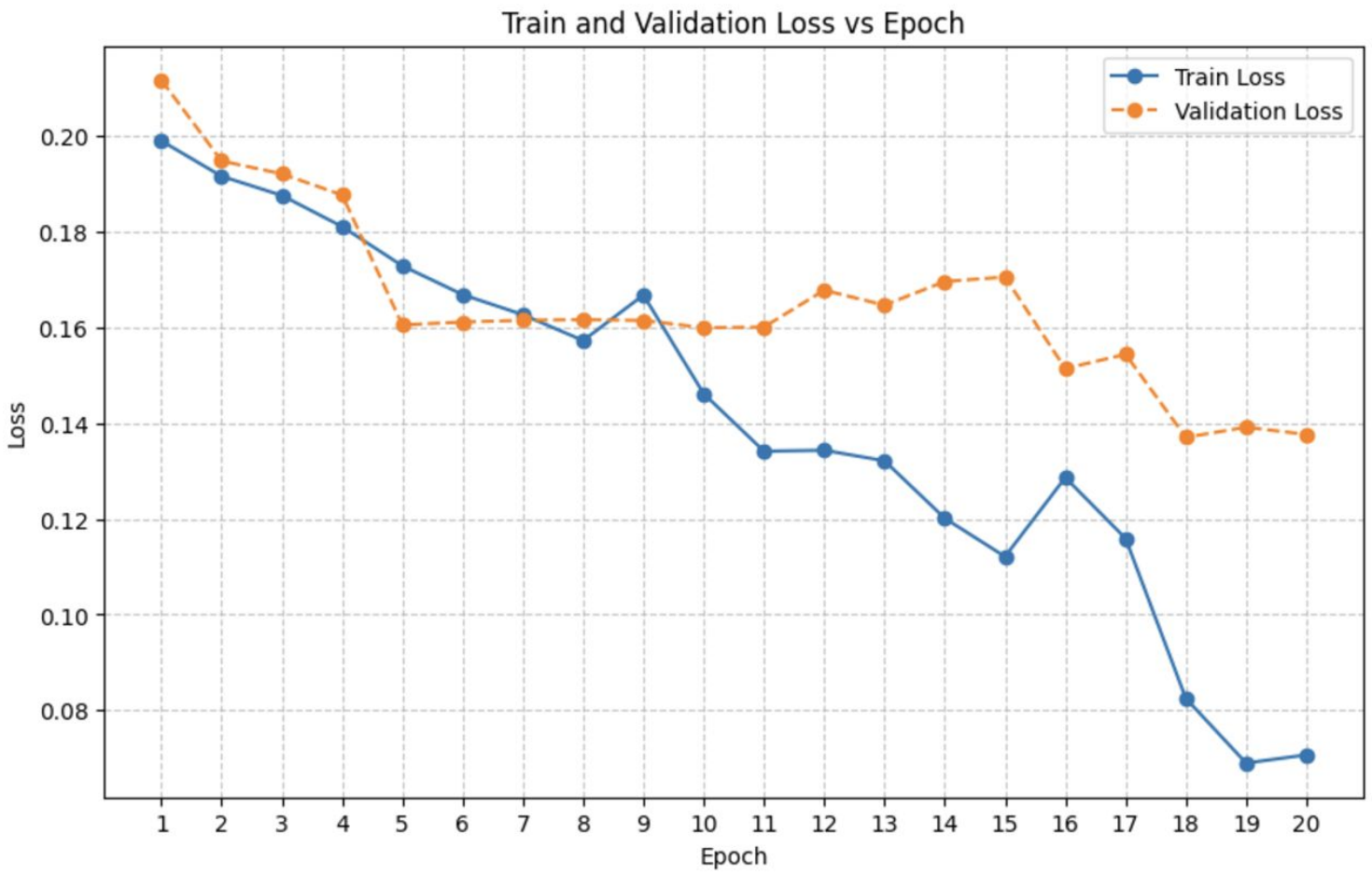
EA-MT

Entity-Aware Machine Translation

EA-MT

1. Process the entity from SpaCy to <<Entity Label: Entity Text>>.
2. Set source language (en_XX) and target language (zh_CN).
3. Used Tokenizer to padding and truncation let max length 256
4. IWSLT Train Data split: 90% for training 10% for validation
5. Used A100 from Google Colab
6. Hyperparameters
    1. Learning rate: 0.00001
       Stability during fine-tuning
    2. Batch size: 4 Prevent GPU memory overflow.
    3. Regularization: Weight decay set to 0.01.
    4. EarlyStoppingCallback: It will stop if there is no progress after 3 epochs.

Training and Validation Loss



Training Loss :

- From 0.1989 (Epoch 1) to 0.0707 (Epoch 20)
- After 10 Epoch model Improved a lot

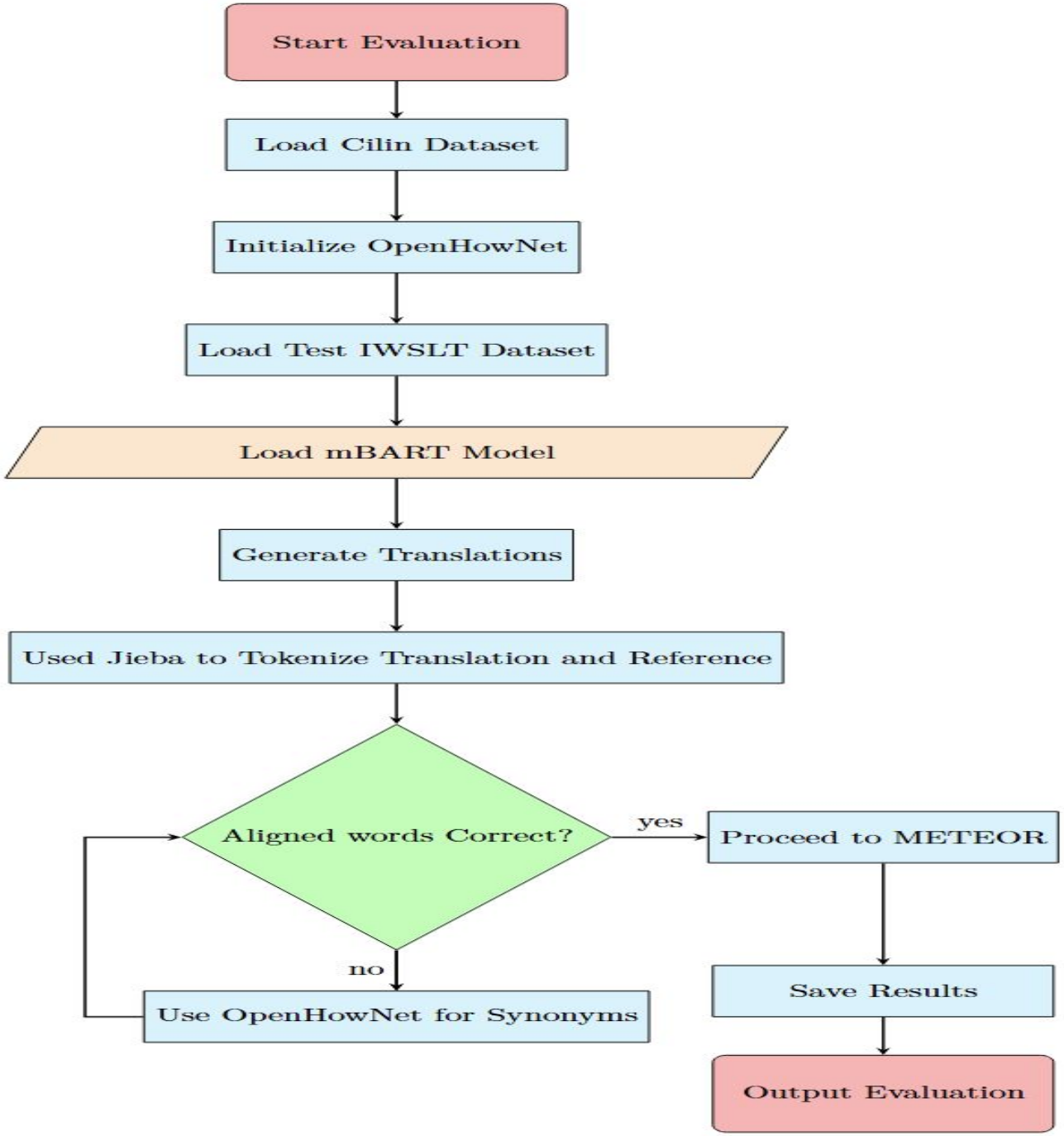
Validation Loss:

- From 0.2116 (Epoch 1) to 0.1376 (Epoch 20)
- Fluctuations starting from Epoch 10 stabilized near 0.14.

PART.07

EA-MT
Evaluation Flowchart

EA-MT
Evaluation Flowchart



PART.08

METEOR & Jieba

Evaluation of Entity-Aware Machine Translation

METEOR

METEOR Score: 0 ~ 1

0 = Completely wrong translation

1 = Same as reference translation

Overall METEOR Score: 0.04137237761331027

==========

"Original Text": "Day four."
"Reference Text": " 第四天。"
"Translated Text": " 第四天。"
METEOR: 1

==========

Original Text: You know, we wake up in the morning and we feel we make decisions.
Reference Text: 我们早上起来，觉得可以主宰自己的决定”
Translated Text: 大家知道,我们早上起来后,我们感觉自己要做决定。
METEOR: 0

==========

Original Text: If they want sad music, they just play those two notes.
Reference Text: 他们就不停的用这两个音符。
Translated Text: 如果他们想要悲伤的音乐,他们就只演奏这两段音符。
METEOR: 0

==========

METEOR

1. **Tokenization Issues:**
   METEOR Eval is for a language that has clear word boundaries.
   Solution: ***Jieba*** is a Chinese segmentation tool.
   (Chinese is character-based)
   English: "This is a book." => ["This", "is", "a", "book"]
   Chinese: "这是一本书" => ["这是一本书"]
   Jieba: "这是一本书" => ["這", "是", "一本", "書"]

2. **Differences in the punctuation (comma) format.**
   Solution: Used regular expressions to delete punctuation.

3. **Lack of Built-in Synonym Support for Chinese**
   METEOR used WordNet for word-similarity matching, not supporting Chinese.
   Harbin Institute of Technology Word-Similarity Dataset (HIT-CIR)

4. **Not Enough Attention in Word-Similarity Dataset**
   Harbin Institute of Technology Word-Similarity Dataset is not enough.
   OpenHowNet.calculate_word_similarity to match similar words.

PART.09

HIT-CIR & OpenHowNet

Word Similarity Dataset & Calculate Word Similarity

HIT-CIR Word-Similarity

HIT-CIR ≈WordNet

Organized into five levels:

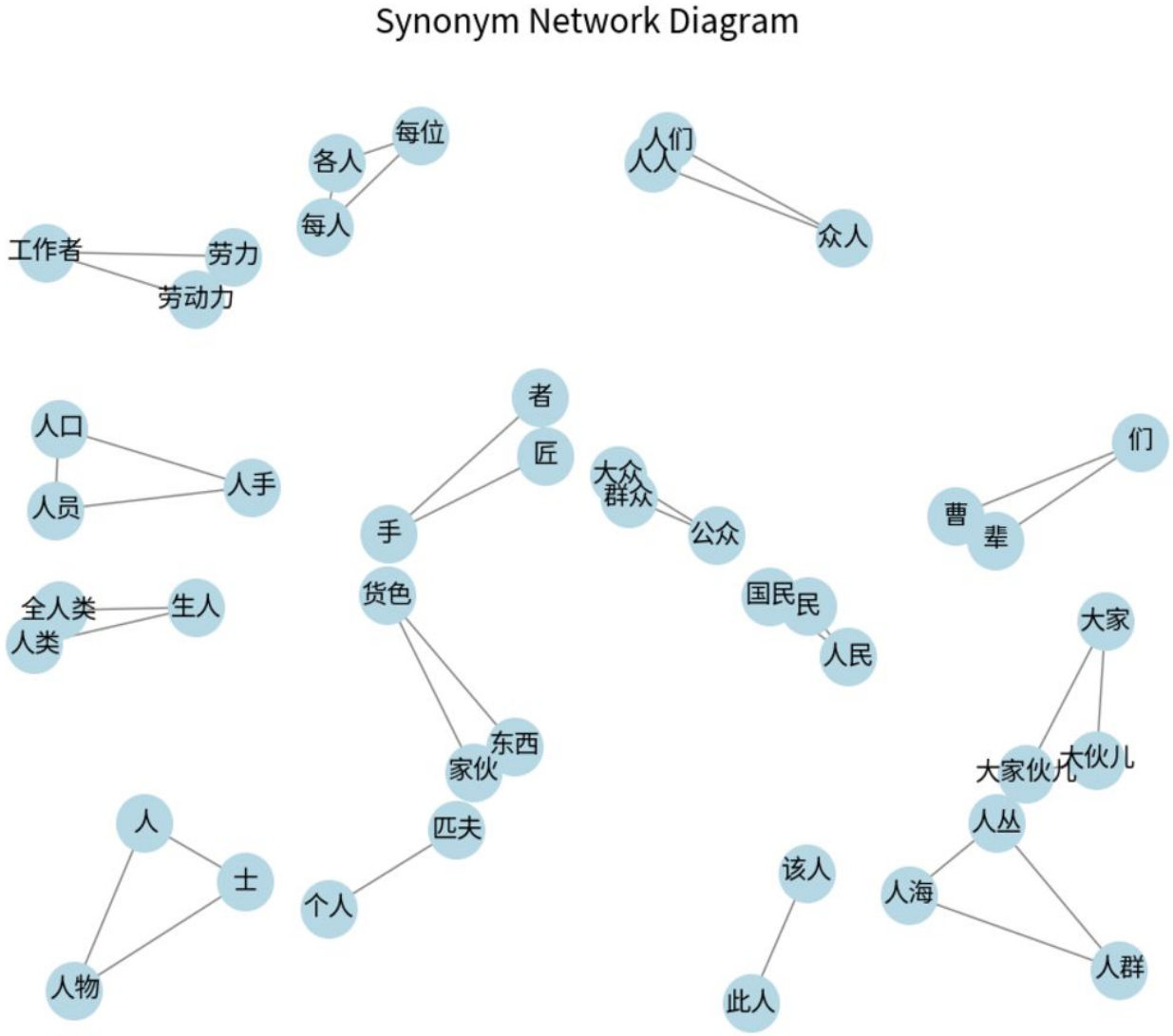
1. Top-level categories ("Person", "Object").
2. Subcategories (e.g., "Human Behavior").
3. Specific subcategories.
4. Semantic units.
5. Individual words or phrases.

OpenHowNet.calculate_word_similarity

0: Not *similar*

1: Completely match

0.7: considered synonyms word



PART.10

Evaluation Results

Entity-Aware Machine Translation Result

Evaluation Results

指標	分數
METEOR	0.6649
BLEU	0.0686
BERTScore	0.8242 (F1)
GLEU	0.0455

Overall METEOR: 0.6649

Reflecting strong semantic and structural alignment.

BERTScore:

Precision: 0.8008

Calculates the percentage of the translated text that accurately matches to the reference.

80.08% of generated translations match the reference translations.

Recall: 0.7933

Calculates the percentage of the reference material that the translated text covers.

79.33% of the reference translation's content is successfully translated.

F1: 0.7965

Reflects a good balance between translation accuracy and semantic coverage.

Language	Mean GLEU	Mean BLEU	Mean chrF	Mean METEOR
Arabic	0.354	0.225	0.559	0.592
Vietnamese	0.292	0.173	0.483	0.526
Chinese	0.155	0.039	0.358	0.318

BLUE: 0.0686

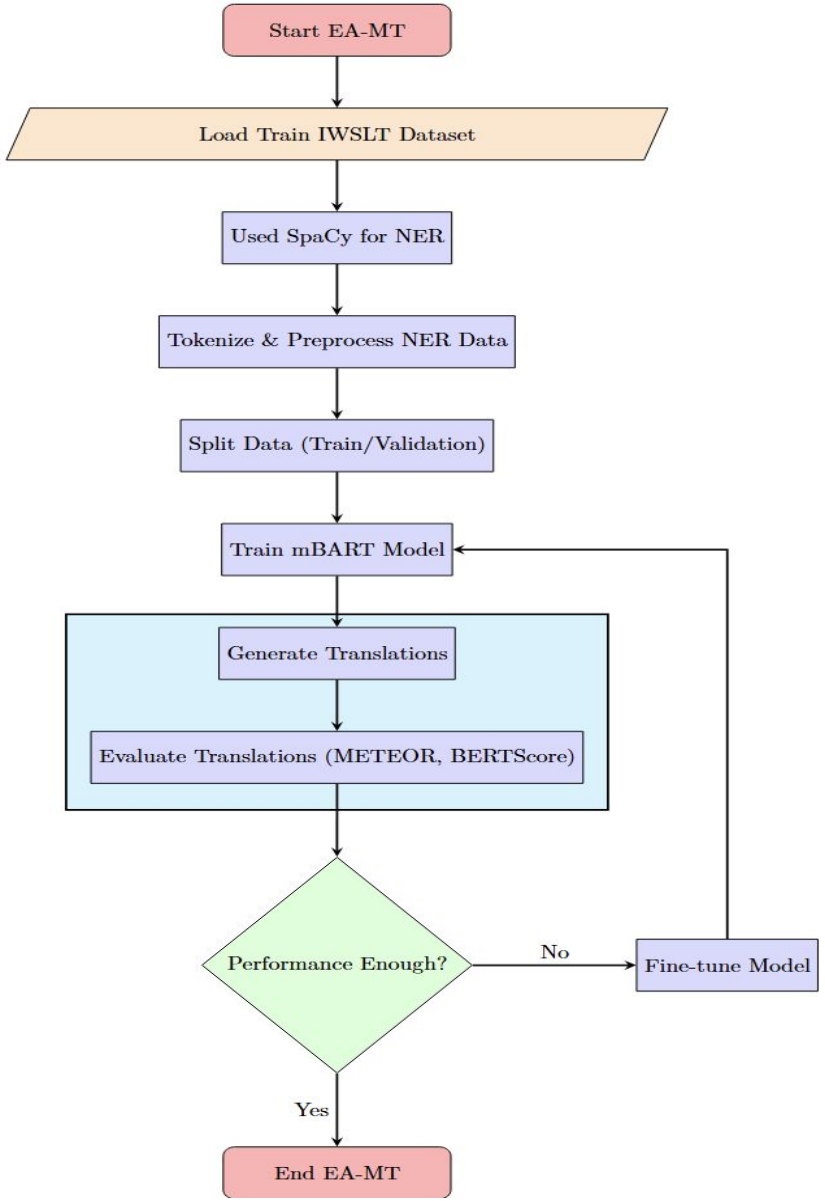
0 : Not matching

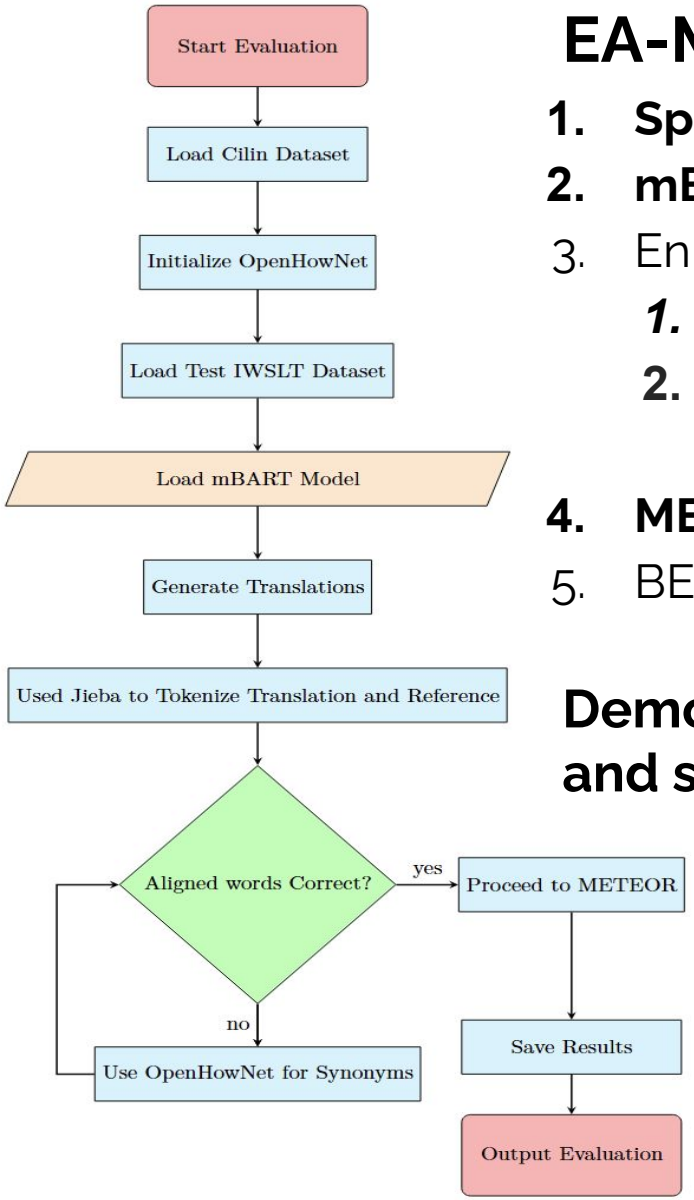
1 : Completely match

Automated Multi-Language to English Machine Translation Using Generative Pre-Trained Transformers

Hypothesis: Chinese grammar structure is relatively different from English.

Conclusion





EA-MT English-to-Chinese system:

1. **SpaCy** for **NER**
2. **mBART** for multilingual machine translation.
3. Enhancing evaluation metrics:
   1. ***Jieba*** Chinese tokenization
   2. **HIT-CIR & *OpenHowNet*** synonym word matching
4. **METEOR** score: 0.6649
5. BERTScore F1: 0.7965

Demonstrating superior translation quality and semantic alignment.

Q
A